

MEC ENG 102B
Mechatronics Design
Project Assignment 1 & 2
Team Kickoff and Pitches

Submit a single PDF with all deliverables from both P1 and P2.

PROJECT ASSIGNMENT 1

One of the most important, and surprisingly challenging, skills to learn is effective teamwork. A key element to success is the willingness to grow and listen to others. These exercises are intended to ease team interactions throughout the semester. In my experience, this level of communication early-on is helpful whether people have worked together before or not, as people's circumstances and values change over time. **The more thoughtfully you take this activity, the more you are likely to benefit from it!**

Part 1: Getting to know your team

Take some time to get to know each other in the context of this class. Each person should answer each of the following questions. Making sure each person has between 5 to 10 minutes uninterrupted to share their responses.

- 1) What major commitments (outside of ME102B) do you have this semester? In this context, how central is the ME102B project for you this term? How are you hoping to engage with it?
- 2) How do you prefer to communicate? Via prescheduled meetings, texts, slack, etc.? Do you enjoy working alongside others or working alone typically? What times do you expect to do your work?
- 3) What skills are you comfortable with that you can bring to your team or teach others? What skills are uncomfortable for you or that you are hoping to develop through this course?
- 4) Any other hopes or concerns about the project, team dynamic, or class this semester?

Part 2: Collaborative Agreement

Create an agreement that you can refer to later in the semester; the hope is that you can face the project with confidence even when unexpected challenges arise. Make the agreement 1 page or less and have everyone on your team sign it. If teaming issues arise later in the semester, the teaching team may refer to this agreement in arbitration. It must include at least these elements (you are welcomed to include more):

- 1) **Shop training:** Who will be shop trained? Will this personal also have access to CAD files for quick modifications? If someone has less prior experience in the shop, how do they get involved?
- 2) **Sourcing parts:** Who will be responsible for purchases, shipping, and returns? Is this something everyone will share or not? The ASME Maker Grant program is reimbursement at the end of the semester. Someone also needs to be the point person if parts are borrowed from Tom.
- 3) **Deadlines:** How far in advance will each person complete work prior to a deadline? Who will be responsible for submitting work – will this be the same person, or will it rotate?
- 4) **Absence, and other teammate concerns:** On occasion, a student stops responding to communications or showing up to meetings. Or maybe they complete work late, inconsistently, or haphazardly. It's important to acknowledge that this is often situational and may indicate that the student needs special accommodation to complete the class. Remember that the only time when teammate evaluations affect grades is at the very end of the semester, so it's best for everyone to address concerns in a timely manner. Generate a *well-defined procedure* that your team can use to address future concerns like these if they arise. For example: How many reminders and in what format do you expect to send an absent teammate before notifying the teaching team of the absence or underperformance? Think creatively about what format makes sense for your team. The teaching team is here to give advice and support – we want everyone to do well this semester.

***Deliverable 1* submit your collaborative agreement.**

PROJECT ASSIGNMENT 2

There are three sections of this assignment: Opportunities, Solutions, and Pitch. These activities must be completed in order and as a group. Note the distinction between Opportunity and Solution; an Opportunity is a need or desired outcome, while the Solution is the specific concept to address that Opportunity. These exercises are designed to help you generate a project that is exciting for you and your team. Putting the effort in now will pay off throughout the rest of the semester as you work together to realize a collective vision. Throughout, you are encouraged to speak with the teaching team if you have questions or concerns.

FYI these are the award categories (selected by neutral panel of judges at the public showcase):

- **Mechanical Design:** For the project with the most elegant or effective use of physics, mechanisms, or mechanical advantage
- **MEECS/Electromechanical Design:** For the project which demonstrates the most sophisticated use of software or electronics to control a system.
- **Design Innovation:** For the most original or unique project idea.
- **Social Impact:** For the project with the greatest potential to address an existing societal challenge, e.g., climate change, inequity, etc.
- **Commercial Value/Entrepreneurship:** For the project with the greatest commercial viability or real-world, practical value.

-- Opportunities --

We will first identify promising *opportunities* for a new mechatronic machine, device or product. First brainstorm problems you try to solve with a device by following the thinking processes explained in the slides below. For example, do you want to create an educational tool for kids, get more energy out of solar panels for a family living on the Berkeley Hill, etc.? *You are not looking for implementation solutions yet.* As a group, select a primary project opportunity. The goal is to converge on an idea that, in genuine ways, excites all team members.

Keep an open mind and encourage participation from all members on the team. Remember:

- During divergent thinking, don't be critical of your own or others' ideas.
- During convergent thinking, consider the underlying values and goals of yourself and the team.

The team must converge on a single opportunity/motivation you will pursue for the rest of this semester.

Refresh from the lecture: Inclusive Design Mindset to generate design ideas:

Mismatch (Who)

Different environments enable different capabilities, present different limitations, and have different rules and social norms.

Adapted from [Microsoft Design](#)

	Permanent	Temporary	Situational
Touch	 One arm	 Arm injury	 New parent
See	 Blind	 Cataract	 Distracted driver
Hear	 Deaf	 Ear infection	 Bartender
Speak	 Non-verbal	 Laryngitis	 Heavy accent

Conditions (When)

Different environmental conditions can change our situational limitations.

Adapted from [Microsoft Design](#)

Weather



Temperature

0°

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Optional Funding Opportunity for Final Project

Mechatronic Design for Natural Ecological Restoration, supported by a National Science Foundation project “Rooted in Nature: Bioinspired Design of Sustainable Seeding Methods to Improve Forest Regeneration”, Led by Prof. Lining Yao. Ecological restoration is the process of assisting the recovery of an ecosystem that has been degraded, damaged, or destroyed. We welcome project proposals that are related to mechatronics design in this context.

- \$300 - \$400 technical material supply support depending on levels of interests
- Aspects that could be relevant to but not limited to:
 - Autonomous Reforestation (robotic planting from sky or land)
 - Biodiversity Surveys (via land, sky or water, could collect diverse types of signals from visual, audio to biomarkers)
 - Precision Habitat Maintenance (Plant care & manipulation, watering, fertilization, rodent prevention, invasive plants removal)
 - Natural Disaster Detection Systems (e.g., Fire Monitoring)
- Example project ideas that could be relevant to but not limited to:
 - VTOL drone for aerial seeding
 - Fire monitoring robot
 - Plant watering robot
 - Seed guard robot
 - On-land seeding bot
 - Exoskeleton assisting human to plant
 - Projectiles for seed dispersal over distance

Deliverable 2: Fill in the following sentence to describe your selected opportunity:

"For [person/more-than-human] who is [in a situation], it's a [exoskeleton] that [action] when [trigger/s]. (opt in/out to be considered by the optional funding opportunities) "

Please ensure your idea falls within the general category of a mechatronic device or machine that includes at least one moving transmission/degree of freedom, one DC motor, one analog input, and one digital input. You are welcome to use multiple sentences if necessary to describe multiple actions or triggers.

Examples:

- For volunteers who have to manually plant seedlings into hard and dry soil for ecological restoration, it is a ‘third arm’ a person can wear to help dig holes and apply water.
- For a stressed-out student who is hyperfixating on studying, it’s a water-carrying backpack that extends a drinking straw every half hour.
- For dogs who suffer from paralysis or spine injuries and are trying to walk or recover, it’s a wearable wheelchair that assists with walking and rehabilitation.

-- Solutions --

After you select an opportunity, move to *solutions*. In particular, we want you to produce 3 different tangible device concepts. You should have at least 3 ideas and no more than 6. We will use these concepts in a decision matrix exercise (weighted matrices).

To perform the weighted matrix to evaluate the concepts, place design options/themes as columns and the design factors/criteria (e.g., feasibility, cost, team enthusiasm, aligned with opportunity, etc.) as rows. Each criterion is assigned a weight to show its relative importance. Each design option is numerically scored for each criterion and this score is multiplied by the criterion's weight. Weighted scores are summed, giving a total score for each design option. This is an especially useful method to converge on ideas as a group when there are several alternatives and varying criteria to consider. It also provides a structured way to decrease the impact of individual biases in the decision-making process.

Follow these steps:

- a) Discuss and refine a list of criteria. Identify any criteria that must be included, focusing on reducing the list of criteria to those that the team believes are most important (approx. 5).
- b) Assign a relative weight to each criterion, based on how important that criterion is to the situation. Do this by distributing 10 points among the criteria. The assignment can be done by discussion and consensus. Or each member can assign weights, then the numbers for each criterion are added for a composite team weighting.
- c) Draw an L-shaped matrix. Write the criteria and their weights as labels along one edge and the list of options along the other edge.
- d) Evaluate each choice against the criteria.
 - Method 1: Establish a numerical rating scale for each criterion, like 1 = little to 5 = great.
 - Method 2: For each criterion, rank-order all options according to how well each meets the criterion. Number them with 1 being the option that is least desirable according to that criterion.
 - Method 3, Pugh matrix: Establish a baseline, which may be one of the alternatives or the current product. For each criterion, rate each other alternative in comparison to the baseline, using scores of worse (−1), same (0), or better (+1). Finer rating scales can be used, such as 2, 1, 0, −1, −2 for a five-point scale.
- e) Multiply each option's rating by the weight. Add the points for each option.

Keep in mind: The option with the highest score after the first try will not necessarily be the one you choose as a group, but the relative scores can generate meaningful discussion and lead the team toward agreement. If you find that you want to change the weights of the different criterion to get a different outcome, this may indicate that you have an important criterion that's misrepresented, or even missing altogether! Remember that respectful disagreement can enhance a device or product by bringing complex objectives to the surface. The teaching team is here to provide advice as you navigate this decision as a team.

Deliverable 3: Show a condensed weighted matrix (1 page max), with the criteria, concepts, and scores shown. State which concept you've selected to pursue and why. Say if you selected a concept that did not score the highest and, if so, why.

Teams must converge on one solution idea prior to moving to the Pitch section.

-- Pitch --

Congrats! You've generated and selected a creative and exciting opportunity and initial solution concept. Now, you'll create a pitch to get feedback from the teaching team.

The following deliverables (4a-d) should be in the form of a PDF export of your slides (4 slides maximum) that you will also use during your pitch presentation with the instructor.

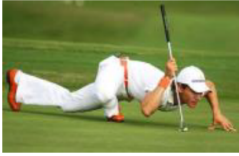
Four deliverables must be included in the pitch. Below find toy examples for a fake project for reference.

Deliverable 4a: Short problem description

Briefly introduce the project *opportunity* (as in Deliverable 2), as well as why your team selected it. Also report if there are any similar solutions that already exist. (Hint: do some digging online).

Hole-in-one-bot (HIO-bot): a speedy ball-shooting robot

- Building a robot that can locate a hole on a putting green and shoot a hole-in-one.
- *Why work on this device?* Because we **love robots and golf** and the whole team is interested in the **dynamic control of a wheeled vehicles**.
- The product is for **entertainment or educational purposes**.
- Existing devices include...



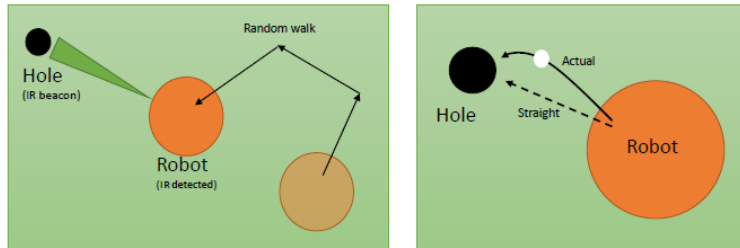
Deliverable 4b: Explain the high-level strategy of your solution concept

Briefly describe the *strategy* that your solution takes. Include a simple cartoon of behavior (e.g. multiple frames) showing how your solution behaves (not mechanism details). Communicate the desired outcome of the mechatronic product clearly and succinctly. It is strongly suggested that this series of figures be made in a program like ppt, inkscape or illustrator and include text labels to make it easy for the teaching team to interpret. If it is not easily readable, we will not be able to give useful feedback on feasibility. You will also be able to use these figures (or updated versions) for your final project deliverables.

One aspect of this class is to generate achievable and measurable objectives. So, keep in mind that the behavioral goals you define may be used to assess the success of your project later. (You will get feedback on feasibility from the teaching team and may update them before final assessment.) Differentiate clearly between essential goals (for the required project elements) and “reach” functionalities that are not necessarily aligned with required class elements. For example, “we will produce the [subassembly] first to satisfy all project requirements. With remaining time in the semester, we will build a second [subassembly] to demonstrate [more complex behavior].”

Strategy of the HIO-bot

- HIO-bot will **drive around on golf greens** (i.e. grassy/sloped surfaces). This green will have **flagsticks with an IR beacons for homing**.
- HIO-bot will start at the edge of the green and **randomly drive** around until it spots the beacon. It will then **stop and align with the hole and shoot a ball**. An observer can tell the bot if it succeeded with **feedback buttons**.
- Special considerations to included **as time permits**: a learned model to help it compensate for slopes when shooting using the data gathered, and wheel redesign that helps them maintain traction on sloped grassy surfaces.



Deliverable 4c: Demonstrate this overall behavior with a low-fidelity mockup (Pictures/video clips in slides + bring this material, as feasible, to your pitch)

Get your hands on any materials that will be integral to your opportunity/solution. If you are making a HIO-bot, then you would need to get some golf balls and go out to a golf green. Maybe get a patch of turf that you will use in your final demo showcase. Get a sense for how much force or velocity you might need to get the ball to roll a certain distance. Etc. Get some cardboard or other prototype materials and mockup rough pieces to experience the concepts/subsystems you're envisioning. For example, make a ball shoot out of a scrap cardboard tube to test whether you can consistently get a hole-in-one by aiming the tube towards the hole and dropping balls down it – how hard is this task really? How far away can you get it to work? Etc. You should play-act the behavior of the machine. On this slide, report your findings with pictures and observations.

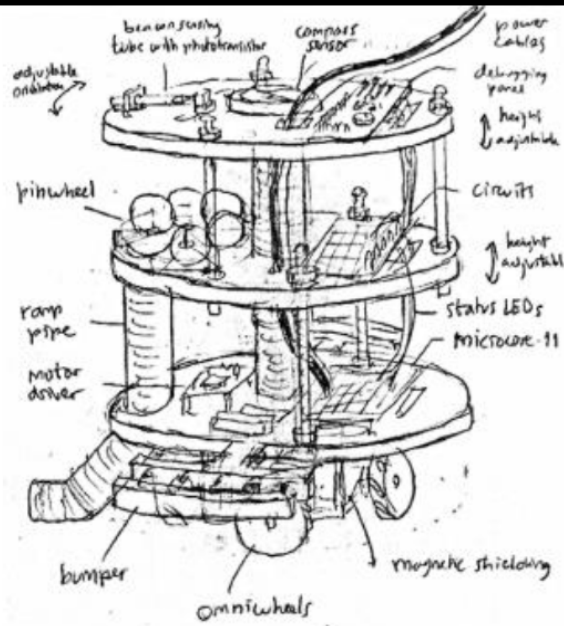


Deliverable 4d: Propose a more detailed system-level design solution

Create a *sketch* (DO NOT USE CAD) **showing the degrees of freedom of your device** (e.g. how does it move?). Show where motors, sensor, electronics and wires are to be mounted and routed and how they connect. **Highlight/label required course elements**. This is a preliminary vision, knowing that it is very likely to change as you start working and get feedback from the instructors. Propose a set of specific **metrics** you wish to achieve (i.e. how will you know when this idea “works”?).

Integrated system

- Omnidirectional wheels enable it to translate as well as to rotate
- An IR sensor detects the IR beacon on the flag indicating hole position
- The balls start up high, so that they have enough velocity when leaving the tube in order to reach a hole xx meters away
- We want the robot to drive up to at least xx m/s
- One goal is to have it battery powered, so it is not tethered



Notice how there are no “floating” or ungrounded parts in the sketch on this slide. It is not super detailed, but you can see how it could all fit together.